Context-Aware Guided Attention Based Cross-Feedback Dense Network for Hyperspectral Image Super-Resolution

Wenqian Dong, *Member, IEEE*, Jiahui Qu, *Member, IEEE*, Tongzhen Zhang, Yunsong Li, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Convolutional neural networks (CNNs) have shown impressive performance in computer vision due to their nonlinearity. Particularly, DenseNet (DN) that facilitates feature reuse in a feedforward (FF) manner has achieved state-of-the-art reconstruction accuracy for super-resolution (SR). However, most DN-based SR models transfer the features generated from each layer to all the subsequent layers, inevitably introducing redundancy, especially for high-dimensional hyperspectral (HS) images. To tackle this problem, we propose a two-branch cross-feedback dense network with context-aware guided attention (CFDcagaNet) for HS super-resolution (HSSR), which allows the network to learn the attention maps of high-level features and refine the low-level features in a feedback (FB) manner across two branches. Context-aware guided attention (CAGA) uses high-level posterior information to provide more faithful spatial–spectral guidance for low-level features, which enables CFDcagaNet to learn more effective spatial–spectral features at low levels and yield more effective spatial–spectral transfer in the network. Extensive experiments on widely used datasets demonstrate that the proposed method outperforms state-of-the-art methods in terms of both quantitative values and visual qualities.**

***Index Terms*— Context-aware guided attention (CAGA), cross-feedback, dense network, hyperspectral (HS) image super-resolution.**

I. Introduction

HYPERSPECTRAL (HS) image is a 3-D data cube containing spatial and spectral information. Since HS images have a high spectral resolution for identifying various materials of interest and achieve the finer classification of land cover categories (as shown in Fig. 1), they are widely used in many applications, such as military, agriculture, and environment [1], [2]. However, due to the limited incident energy and the tradeoff between signal-to-noise ratio, spatial resolution, and spectral resolution, HS image has high spectral resolution but low spatial resolution, which greatly limits its application. HS pansharpening, which is a form of HS super-resolution (HSSR), aims to fuse a low-resolution HS (LR-HS) image with a high-resolution panchromatic (PAN) from the same scene to generate a pansharpened HS with higher spatial resolution [3], [4]. Due to its fine result, HSSR has been widely applied in applications such as land-cover and land-use classification [5], [6], target detection [7], and anomaly detection [8], [9]. Fig. 2 shows an application example of our proposed cross-feedback dense network with context-aware guided attention (CFDcagaNet) in anomaly detection. Experiments show that, compared with the original HS image, performing anomaly detection on the pansharpened HS image can indeed significantly improve the detection accuracy. Reference [10] also shows that pansharpening is beneficial to improve the performance of anomaly detection.

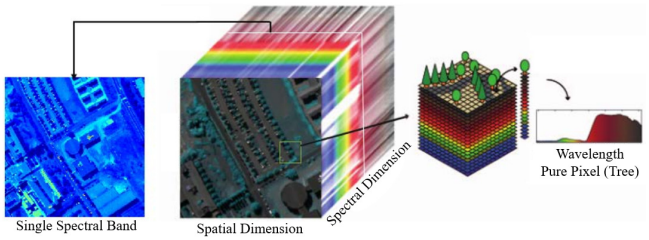


Fig. 1. Schematic illustration of HS image.

Over the past decades, a great number of fusion-based HSSR algorithms have been proposed, and they can be roughly divided into four categories: composition substitution (CS) methods, multiresolution analysis (MRA) methods, matrix decomposition methods, and Bayesian-based methods [11]. The CS-based methods [12]–[14] replace the spatial components of an original HS image with a high spatial resolution PAN image and reconstruct the HS image through inverse transformation. However, due to the mismatch between the spectral range of the PAN image and the HS image, these methods suffer from severe spectral distortion.

Manuscript received March 9, 2022; revised April 28, 2022 and June 1, 2022; accepted June 2, 2022. Date of publication June 6, 2022; date of current version June 16, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 62101414, in part by the Natural Science Basic Research Plan in Shaanxi Province of China under Grant 2021JQ-194 and Grant 2021JQ-197, in part by the Fundamental Research Funds for the Central Universities under Grant XJS210108 and Grant XJS210104, in part by the China Post-Doctoral Science Foundation under Grant 2021M702546 and 2021M702548, in part by the Scientific and Technological Activities for Overseas Students of Shaanxi Province under Grant 2020-017, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2020A1515110856. *(Corresponding author: Jiahui Qu.)*

Wenqian Dong, Jiahui Qu, Tongzhen Zhang, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: wqdong@xidian.edu.cn; jhqu@xidian.edu.cn; tzzhang_1@stu.xidian.edu.cn; ysli@mail.xidian.edu.cn).

Qian Du is with the Department of Electronic and Computer Engineering, Mississippi State University, Starkville, MS 39759 USA (e-mail: du@ece.msstate.edu).

Digital Object Identifier 10.1109/TGRS.2022.3180484

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

The MRA-based methods [15], [16] inject the high-frequency component of the PAN image into resampled HS image to obtain high-resolution HS (HR-HS) images. Although these MRA-based approaches can efficiently reduce the spectral distortion, they usually cause spatial distortions, such as ringing artifacts, due to the mixing of low-frequency components. The matrix decomposition methods [17] alternately decompose high-resolution PAN and HS images to obtain accurate end-members and abundance maps to reconstruct HS images. The Bayesian approach [18]–[20] relies heavily on the posterior distribution of the desired HS image. They have achieved satisfactory SR results, but the solid prior information is the basis of these methods to achieve excellent results, resulting in limitations in practical applications.

In recent years, deep learning has drawn much attention in computer vision due to its strong feature representation capability [21]. A taxonomy of deep-learning-based remote sensing data analysis methods can be found in the literature [22]. Wei *et al.* [23] introduced the concept of residual learning into a very deep CNN for a better multispectral SR performance. In [24], a joint spatial channel attention module is adopted to a differential information residual CNN, which effectively improves the ability of the network to extract features. Reference [25] developed an encoder attention module for HSSR, which is embedded in each layer of the feature extraction network to better utilize the spectral and spatial information of PAN and HS images. He *et al.* [26] proposed a novel CNN-based HSSR architecture, which introduces a spectral predictive structure to enhance the spectral prediction ability. Zhang *et al.* [27] presented an unsupervised adaptation learning (UAL) framework for HSSR, which benefits from implicitly learning using deep networks. Dong *et al.* [28] developed a Laplacian pyramid dense network for fusion-based HSSR, which applies the mature Laplace pyramid decomposition technique to the CNN to simplify the HSSR problem. In order to further improve the HSSR performance, they presented a fusion-based HSSR framework based on a generative dual-adversarial network, in which the HSSR problem is formulated as a dual-task and solved by a generative adversarial network with two discriminators [29]. Wang *et al.* [30] proposed a content-aware text super-resolution (SR) network, which designs a novel text perceptual loss to make the generator aware of the content of the text and produce recognition-friendly information. He *et al.* [31] developed an optimization-driven network for spectral SR by unfolding a variational method. Xue *et al.* [32] proposed a structured sparse low-rank representation model to make full use of the spatial/spectral subspace low-rank relationships between the desired HS image and the observed images to improve the HSSR performance. Dong *et al.* [33] proposed a model-guided deep HSSR network that facilitates an easy intuitive observation on what happens inside the network. Ma *et al.* [34] designed an unsupervised generative adversarial network-based pansharpening framework. Qu *et al.* [35] proposed an unsupervised pansharpening (UP) method based on the self-attention mechanism, which increases the reconstruction accuracy and reduces the spectral distortion. Qu *et al.* [36] further proposed an unregistered and unsupervised mutual Dirichlet-Net (u2-MDN), which solves the HSSR problem without multimodality registration.

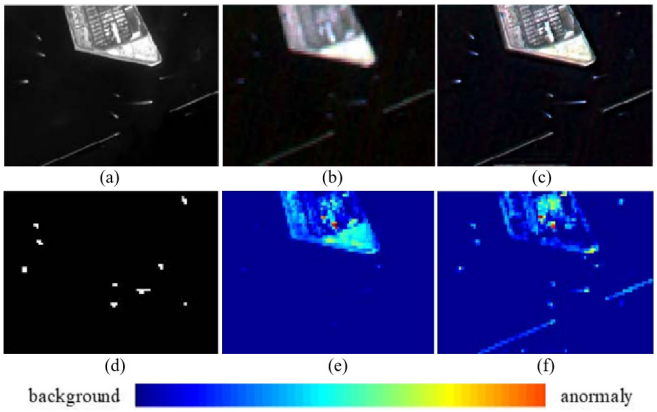


Fig. 2. Application example of our proposed CFDcagaNet in anomaly detection on the Hyperion dataset. (a) PAN image. (b) Available HS image. (c) Pansharpened HS image. (d) Ground truth. (e) Detection map (DM) by the available HS image. (f) DM by the pansharpened HS image.

It is crucial to make full use of the hierarchical features between PAN and HS branches for fusion-based HSSR. DenseNet (DN) is capable of exploiting the hierarchical features by concatenating all feature outputs in a feedforward (FF) manner [37]. An intuitive way is to extend the definition of dense connectivity to a fusion-based HSSR problem. Each image has a branch, and dense connections are added between the pairs of layers within the same branch and between those across different branches. This scheme has been proven effective in many computer vision tasks, such as face image hallucination [38] and real-world image SR [39]. However, each layer of DN directly receives the feature maps of all the preceding layers as inputs in a FF manner, which inevitably introduces redundancy. Redundant features may cause the reconstructed image to be overly blurred or oversharpened. In addition, redundant features will also bring a huge amount of calculation, resulting in a large number of model parameters.

In order to solve this problem, we develop a cross-feedback refinement mechanism that utilizes high-layer features to refine low-layer features, thereby yielding more effective spatial–spectral features transfer within the network. Benefitted from the posterior information in deep layers, more discriminative features can be learned in shallow layers of the network. In order to explore the impact of the feedback (FB) refinement mechanism and its application to fusion-based HSSR, in this article, we propose a two-branch CFDcagaNet for HSSR. Motivated by the superior performance of DN, CFDcagaNet consists of two branches with cross-feedback dense connection to fully extract spatial and spectral features of the PAN and HS images separately. The significant information of low-level features is highlighted so that the more effective spatial–spectral information can be transferred through dense connection. The main contributions can be summarized as follows.

1) HS and PAN images are jointly processed through a two-branch cross-feedback dense network that fully exploits

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

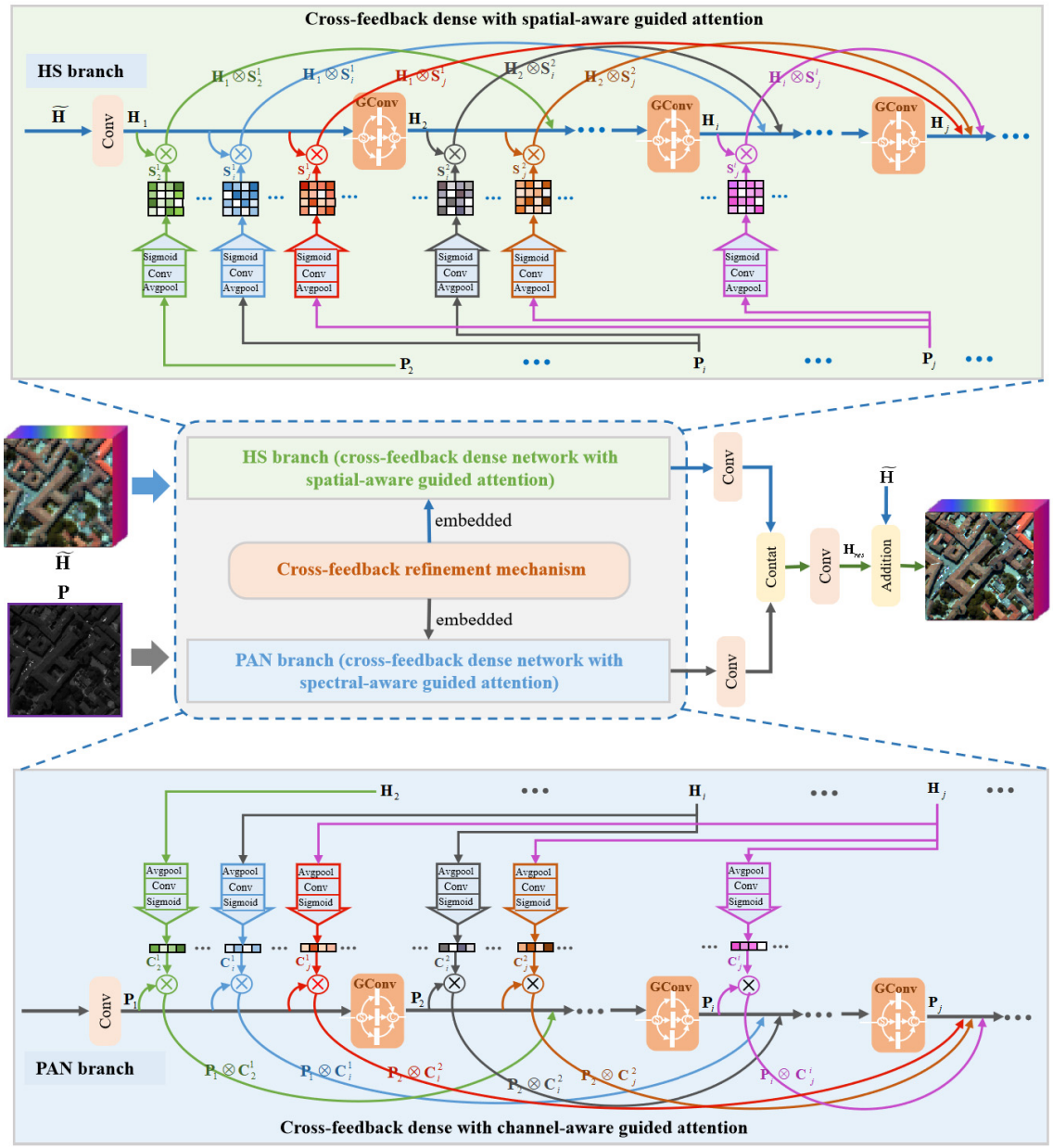


Fig. 3. Architecture of the two-branch CFDcagaNet for fusion-based HSSR. $\{\mathbf{H}_l\}_{l=1,2,\ldots,i,\ldots,j,\ldots}(j > i > 1)$ and $\{\mathbf{P}_l\}_{l=1,2,\ldots,i,\ldots,j,\ldots}(j > i > 1)$ represent the output features of the GConv operation on the HS branch and the PAN branch, respectively.

all hierarchical features within and in between the two branches, enabling better detail extraction and spectral preservation.

2) We design an effective cross-feedback refinement mechanism with a CAGA module, which can provide more faithful spatial–spectral guidance across branches to achieve effective spatial–spectral transfer in the network.

The rest of the article is organized as follows. Section II presents the proposed CFDcagaNet. Section III shows experimental results. Finally, the conclusion is drawn in Section V.

II. Proposed Method

A. Motivation

It is widely accepted that, as the network depth increases, convolutional layers generate hierarchical features with different receptive fields. Suppose that we use N hidden convolution layers, each of which has a receptive field size of $k \times k$. In theory, the output of an input image through i convolution layers with a receptive field size of $k \times k$ is equivalent to the output of this input image through one convolutional layer with a receptive field of $r_i \times r_i$, where $r_i = [(r_{i-1} - 1)s + k]$, s denotes the stride, and r_{i-1} represents the width of the receptive field of the $(i-1)$th convolution layer. The low-level features with small receptive fields focus on capturing local detail information of the targets, while high-level features with large receptive fields can provide global semantic information of the image.

In order to fully extract the effective hierarchical features from different convolutional layers, an intuitive way is to construct a network with two dense branches, one for each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

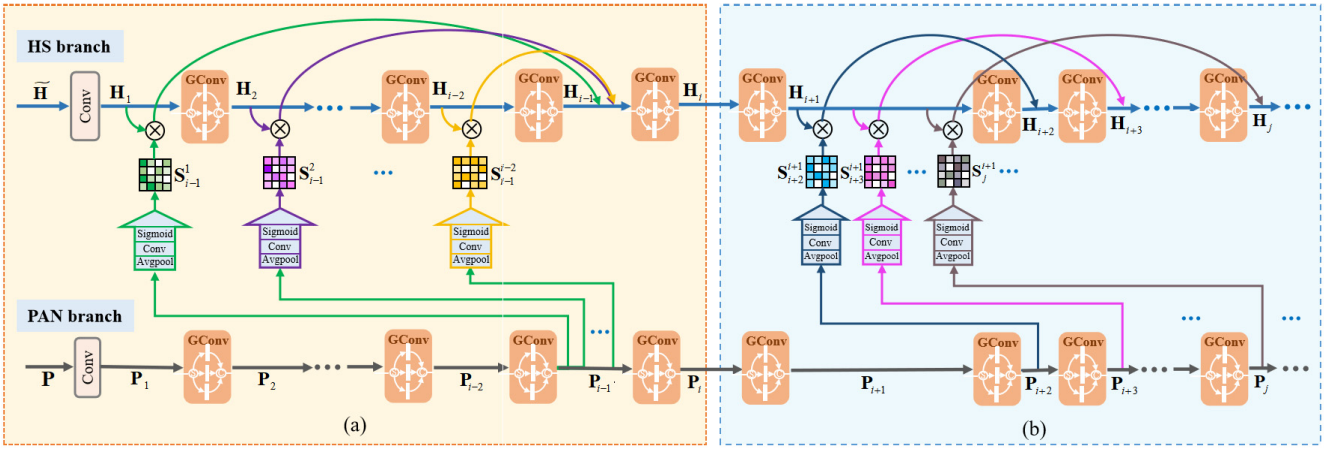


Fig. 4. (a) Schematic illustration of the ith GConv layer of the HS branch in the proposed CFDcagaNet. (b) Schematic illustration of the cross-feedback SpaGA refinement at the $(i+1)$th layer on the HS branch.

imaging modality, where dense connections occur between the pairs of layers within the same branch to extract spatial and spectral information of PAN and HS images, respectively. However, this poses two new problems. On the one hand, it is difficult for such a network to learn more complex combinations between HS and PAN branches. On the other hand, DN works in a FF manner to fuse the shallow local information with the deep semantic information. This layer-by-layer connectivity pattern facilitates feature reuse but also introduces redundancy. How to improve the efficiency of the network is worthy of further study.

To tackle these problems, we propose CFDcagaNet, which learns the attention maps from high-level features of one modality, and use the learned attention map to refine low-level features of another modality in a FB manner. In this way, the high-level semantic features provide faithful spatial–spectral guidance for the low-level features through the cross-feedback refinement mechanism, and the refined low-level features are able to capture more discriminative information, improving the significant flow in the whole network and making network training easier.

B. Problem Formulation

The objective of HSSR is to use HS and PAN images to fully estimate the missing spatial details of HS images. The idea of the residual learning is particularly applicable to solve the HSSR tasks because the available LR-HS image is highly similar to the desired pansharpened HR-HS image, allowing residual learning to learn residual maps with values that may be zero or very small. Based on the above analysis, the proposed CFDcagaNet introduces residual learning to the HSSR task to reduce the unnecessary computational expenses. Given the LR-HS image $\mathbf{H} \in \mathbb{R}^{w \times h \times C}$ and the PAN image $\mathbf{P} \in \mathbb{R}^{W \times H}$, where w (W) and h (H) are the weight and height of the LR-HS (PAN) image, respectively, and C denotes the number of bands, CFDcagaNet permits the network to learn the residual map between the target HR-HS image and the upsampled HS image and follows the general idea of MRA to inject the learned residual map into the upsampled LR-HS image, which can be formulated as

$$\widehat{\mathbf{H}} = \widetilde{\mathbf{H}} + \mathbf{H}_{\text{res}} \tag{1}$$

where $\widehat{\mathbf{H}}$ denotes the target HR-HS image with a tensor of size $W \times H \times C$, $\widetilde{\mathbf{H}} \in \mathbb{R}^{W \times H \times C}$ is the upsampled LR-HS image, which has been preinterpolated to the same spatial scale of the PAN image, and $\mathbf{H}_{\text{res}} \in \mathbb{R}^{W \times H \times C}$ represents the residual map to be estimated by CFDcagaNet, which can be formulated as

$$\mathbf{H}_{\text{res}} = f_{\text{CFDcagaNet}}(\mathbf{P}, \widetilde{\mathbf{H}}) \tag{2}$$

where $f_{\text{CFDcagaNet}}(\cdot)$ is the function of the whole CFDcagaNet.

C. Cross-Feedback Dense Network

DN is widely used due to its superior performance in image SR. Inspired by the architecture of DN, we propose a two-branch CFDcagaNet for HSSR (as shown in Fig. 3), which refines the features in a cross-feedback manner and transfers the refined features to all the subsequent layers. In this way, the dense connectivity facilitates the refined feature reuse in the network. The CAGA refines low-level features under the guidance of high-level features. The low-level features refined by semantic information are passed on more effective spatial–spectral information to all the subsequent layers, improving the efficiency of the proposed network.

The attention maps generated from high-level features highlight the significant information of low-level features, yielding more effective spatial–spectral transfer in the network. Therefore, each layer of CFDcagaNet takes all previous refinement features as input and refines its own features through higher level attention maps and feeds them to the corresponding subsequent layers. Furthermore, in order to further reduce the network parameters, group convolution (GConv) [40] is utilized to the CFDcagaNet. GConv splits the input features into disjoint groups according to their original source, and each group only performs a group-specific set of filters, reducing the computational cost of the network. For the two branches, each GConv layer consists of a batch normalization (BN) layer to accelerate the training process, a rectified linear unit (ReLU) for extraction of nonlinearities, and a GConv layer. The kernel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

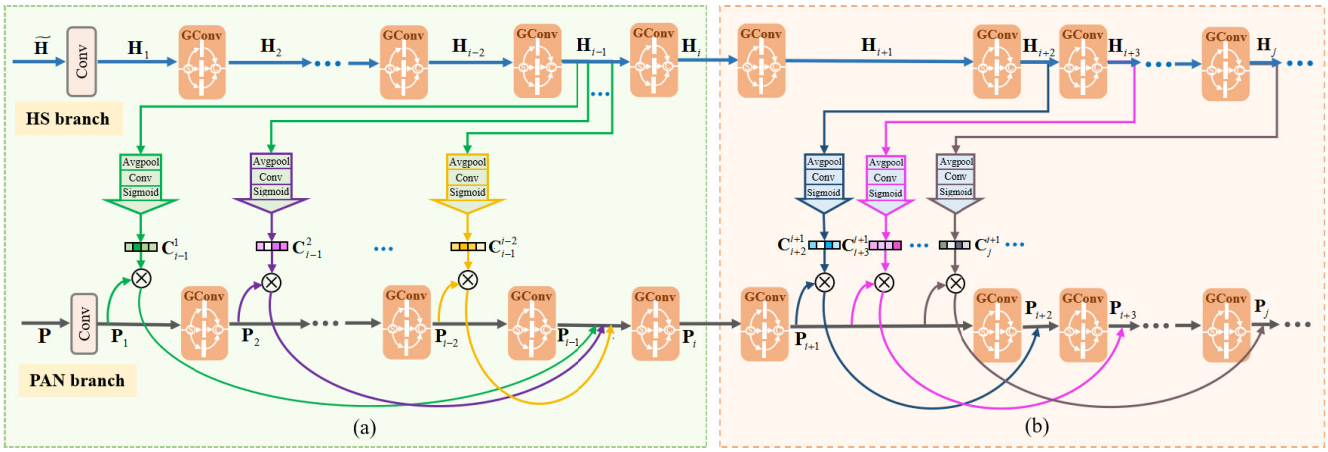


Fig. 5. (a) Schematic illustration of the ith GConv layer of the PAN branch in the proposed CFDcagaNet. (b) Schematic illustration of the cross-feedback SpeGA refinement at the $(i+1)$th layer on the PAN branch.

size of each GConv layer is set to 3×3. As shown in Fig. 4, for the ith GConv layer of the HS branch, the output feature map $\mathbf{H}_i$ can be represented as

$$\begin{aligned}\mathbf{H}_i &= \mathrm{GConv}\big(\mathrm{Relu}\big(\big[\mathbf{H}_{i-1}, \mathbf{H}_1 \otimes \mathbf{S}_{i-1}^1, \mathbf{H}_2 \otimes \mathbf{S}_{i-1}^2, \ldots, \\ &\qquad \mathbf{H}_{i-2} \otimes \mathbf{S}_{i-1}^{i-2}\big]\big)\big) \\ &= \mathrm{GConv}\big(\mathrm{Relu}\big(\big[\mathbf{H}_{i-1}, \mathbf{Hs}_1^{i-1}, \mathbf{Hs}_2^{i-1}, \ldots, \mathbf{Hs}_{i-2}^{i-1}\big]\big)\big) \end{aligned} \quad (3)$$

where $\mathrm{GConv}(\cdot)$ denotes the GConv operation, $\mathrm{Relu}(\cdot)$ is the ReLU activation function, $[\cdot]$ represents the concatenation operation, $\otimes$ denotes the elementwise multiplication, $\mathbf{S}_{i-1}^{i-2}$ is the attention map generated from the features of the $(i-1)$th layer of the PAN branch and used to refine the features of the $(i-2)$th layer of the HS branch, and $\mathbf{Hs}_{i-2}^{i-1}$ is the features of the $(i-2)$th layer of the HS branch refined by the attention maps generated by the $(i-1)$th layer's features of the PAN branch. As shown in Fig. 5, similar to the HS branch, the output feature map $\mathbf{P}_i$ of the ith convolutional layer of the PAN branch can be formulated as

$$\begin{aligned}\mathbf{P}_i &= \mathrm{Conv}\big(\mathrm{Relu}\big(\big[\mathbf{P}_{i-1}, \mathbf{P}_1 \otimes \mathbf{C}_{i-1}^1, \mathbf{P}_2 \otimes \mathbf{C}_{i-1}^2, \ldots, \\ &\qquad \mathbf{P}_{i-2} \otimes \mathbf{C}_{i-1}^{i-2}\big]\big)\big) \\ &= \mathrm{Conv}\big(\mathrm{Relu}\big(\big[\mathbf{P}_{i-1}, \mathbf{Pc}_1^{i-1}, \mathbf{Pc}_2^{i-1}, \ldots, \mathbf{Pc}_{i-2}^{i-1}\big]\big)\big) \end{aligned} \quad (4)$$

where $\mathbf{C}_{i-1}^{i-2}$ represents the attention map generated by the $(i-1)$th layer's features of the HS branch to guide the spectral features of the $(i-2)$th layer of the PAN branch, and $\mathbf{Pc}_{i-2}^{i-1}$ represents the features of the $(i-2)$th layer of the PAN branch refined by the attention maps generated by the $(i-1)$th layer's features of the HS branch.

D. Context-Aware Guided Attention

The CAGA, which consists of the spatial-aware guided attention (SpaGA) and spectral-aware guided attention (SpeGA), is embedded to provide more faithful guidance for feature extraction across branches (as shown in Fig. 3). With the help of SpaGA, the high-level features of the PAN branch are used to guide the feature extraction of the HS branch to pay more attention to more important regions. Similarly, the SpeGA is conducted on each layer of the HS branch to guide spectral information of all the preceding layers of the PAN branch, focusing on more discriminative channels and suppressing unnecessary ones. The SpaGA and SpeGA adopted in this article are detailed as follows.

As demonstrated in Fig. 4(a), the attention maps generated by the $(i-1)$th features of the PAN branch are multiplied by all the $i-2$ preceding features of the HS branch to refine the low-level features. Given the $(i-1)$th feature $\mathbf{P}_{i-1} \in \mathbb{R}^{W \times H \times c}$ of the PAN branch as input, the average pooling is first adopted to aggregate the information along the channel dimension. The generated spatial context descriptor is fed into the 3×3 convolutional layer, and the sigmoid function is adopted to normalize the attention score between 0 and 1. For the $(i-1)$th convolutional layer of the PAN branch, the attention map $\mathbf{S}_{i-1}^m \in \mathbb{R}^{W \times H}$ that is multiplied by the mth feature map can be formulated as

$$\mathbf{S}_{i-1}^m = \sigma_s\big(\overline{\mathbf{W}}_{i-1}^m * \mathrm{Avg}_s(\mathbf{P}_{i-1})\big), \quad m = 1, \ldots, i-2 \quad (5)$$

where Avg_s and σ_s denote the average pooling operation and the sigmoid function in SpaGA, respectively, $\overline{\mathbf{W}}_{i-1}^m$ is the kernel of the convolutional layer in SpaGA, and $*$ denotes the convolution operation. The mth $(m < i)$ feature map of the HS branch can be refined by the attention map of $(i-1)$th layer of the PAN branch. The refined feature $\mathbf{Hs}_m^{i-1} \in \mathbb{R}^{W \times H \times c}$ of the mth layer of the HS branch is calculated by multiplying the feature map $\mathbf{H}_m$ with the attention map of the $(i-1)$th layer of the PAN branch, which can be expressed as

$$\mathbf{Hs}_m^{i-1} = \mathbf{H}_m \otimes \mathbf{S}_{i-1}^m, \quad m = 1, \ldots, i-2 \quad (6)$$

where $\otimes$ denotes the elementwise multiplication.

Fig. 4(b) shows that the features of the $(i+1)$th layer of the HS branch $\mathbf{H}_{i+1} \in \mathbb{R}^{W \times H \times c}$ are spatially guided by all the high-level features after the $(i+1)$th layer of the PAN branch for feature refinement. The refined features are densely passed to the subsequent convolutional layer of the HS branch. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DESCRIPTION OF THE EXPERIMENTAL DATASET (K1: NUMBER OF NONOVERLAPPING PATCHES AND K2: NUMBER OF SMALL PATCHES IN EACH NONOVERLAPPING PATCH)

Dataset	Instrument	Band (Original; Experimental)	Wavelength(nm)	Dimensions (Original; Experimental)	K1	K2	Patches for testing
Houston	ITRES-CASI	(144; 144)	380-1050	(349×1905; 320×1280)	8	41	5, 8
Botswana	Hyperion	(242; 145)	400-2500	(1476×256; 1200×240)	8	21	7, 8
Pavia Center	ROSIS	(115; 102)	430-860	(1096×1096; 1096×640)	12	21	4, 7, 11

process can be expressed as

$$\mathbf{S}_n^{i+1} = \sigma_s\left(\overline{\mathbf{W}}_n^{i+1} * \mathrm{Avg}_s(\mathbf{P}_n)\right), \quad n = i+2, i+3, \ldots \tag{7-1}$$

$$\mathbf{Hs}_{i+1}^n = \mathbf{H}_{i+1} \otimes \mathbf{S}_n^{i+1}, \quad n = i+2, i+3, \ldots \tag{7-2}$$

The SpeGA is built to take advantage of the spectral information of the HS image. Similar to the SpaGA, SpeGA is applied to each layer of the HS branch, and the attention map with high-level posterior information is used to provide more faithful spectral guidance to all the preceding features of the PAN branch. As shown in Fig. 5(a), we use an average pooling layer to squeeze the spatial dimension of the input feature map $\mathbf{H}_{i-1} \in \mathbb{R}^{W \times H \times c}$ of the HS branch. Then, a convolutional layer of kernel size 1×7 followed by a sigmoid function is used to predict the attention map. In addition, the attention map can also be predicted by using two fully connected layers followed by a sigmoid function. However, two fully connected layers will bring greater model complexity. Therefore, we choose 1-D convolution operation to replace two fully connected layers in SpeGA. The attention map $\mathbf{C}_{i-1}^m \in \mathbb{R}^{1 \times c}$ in the $(i-1)$th convolutional layer of the HS branch used to guide the mth feature map of the PAN branch is given as

$$\mathbf{C}_{i-1}^m = \sigma_c\left(\mathbf{W}_{i-1}^m * \mathrm{Avg}_c(\mathbf{H}_{i-1})\right), \quad m = 1, \ldots, i-2 \tag{8}$$

where $\mathbf{W}_i^m$ is the kernel size of convolutional layers in SpeGA, and Avg_c and σ_c denote the average pooling operation and the sigmoid function in SpeGA, respectively.

The SpeGA takes high-level features into consideration to generate better attention prediction in a FB manner. To be specific, the attention map $\mathbf{C}_{i-1}^m$ is used to refine the feature map of the mth ($m < i$) layer of the PAN branch, resulting in more effective representation with redundant information being reduced. The refined feature map $\mathbf{Pc}_{i-1}^m$ of the PAN branch is defined as

$$\mathbf{Pc}_m^{i-1} = \mathbf{P}_m \otimes \mathbf{C}_{i-1}^m, \quad m = 1, \ldots, i-2. \tag{9}$$

Through the above process, we can use CFDcagaNet to more accurately approximate the residual image $\mathbf{H}_{\mathrm{res}}$ in (1).

E. Network Training

The loss function plays a vital role in the performance of the network. Most of the previous HSSR tasks adopt L2 loss as the loss function, while it has been shown that L2 loss is insensitive to small differences. This drawback leads to poor performance in recovering subtle details of the HR-HS image. L1 loss is considered as a better choice for the SR task, and the whole network can be trained in an end-to-end manner [41]. The parameter θ of the network is learned by minimizing the L1 loss between the high-resolution image and the ground-truth image. Given the training set $\{\mathbf{H}^{(n)}, \mathbf{P}^{(n)}, \mathbf{R}^{(n)}\}_{n=1}^N$, where $\mathbf{H}^{(n)}$ and $\mathbf{P}^{(n)}$ refer to the nth training sample of the HS and PAN images, respectively, $\mathbf{R}^{(n)}$ represents the nth training sample of the ground-truth image, and n is the number of training samples, the loss function of the proposed network can be expressed as

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{n=1}^{N}\left\| f(\mathbf{H}^{(n)}, \mathbf{P}^{(n)}; \theta) - \mathbf{R}^{(n)} \right\|_1. \tag{10}$$

III. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we first introduce the experimental settings and the datasets used in the experiments. Then, we evaluate our method with an ablation study. Comparisons with the state-of-the-art fusion-based HS SR methods are given at last.

A. Dataset and Experimental Setting

To evaluate the performance of the proposed method, three publicly HS datasets are considered, including the Houston dataset,[1] the Botswana dataset,[2] and the Pavia Center dataset.[3] Generally, we obtain the simulated LR-HS and PAN images from the reference HS image according to Wald's protocol [42]. Specifically, the simulated LR-HS image is obtained by four times downsampling of the spatial dimension on the result of 9×9 Gaussian blurring of the reference image. The simulated high spatial resolution PAN image is generated by averaging the bands of the HS image in the visible region. Descriptions of the three datasets are summarized in Table I. Note that, in order to increase the training samples, we cropped the simulated PAN and LR-HS images into $K1$ nonoverlapping patches with the same size and further divided each nonoverlapping patch into $K2$ smaller patches at a fixed stride.

All experiments are conducted with the PyTorch framework on the NVIDIA RTX 3090 GPU. The Adam optimizer is used to train the CFDcagaNet with a learning rate of 1×10^{-3}. In addition, the batch size is set to 16.

[1] http://dase.grss-ieee.org/
[2] https://rslab.ut.ac.ir/data
[3] https://rslab.ut.ac.ir/data

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

TABLE II
AVERAGE QUANTITATIVE RESULTS OF THE DIFFERENT MODELS ON THE PAVIA DATASET

Models	Feedforward	Feedback	Group Conv	FLOPS(G)	PARAMS(M)	PSNR	CC	SAM	RMSE	ERGAS	TIME(s)
DN	×	×	×	21.07	0.8233	35.1362	0.9755	4.4885	0.0177	2.7713	0.0093
DN_gc	×	×	✓	**17.49**	**0.6833**	35.0839	0.9759	4.1448	0.0179	2.7697	**0.0085**
FF	✓	×	×	21.09	0.8259	36.2026	0.9801	4.0316	0.0156	2.4366	0.0096
FF_gc	✓	×	✓	17.50	0.6859	36.2950	0.9802	3.9658	0.0155	2.4416	0.0091
FB	×	✓	×	21.09	0.8259	36.5498	0.9815	3.8278	0.0150	2.3237	0.0094
FB_gc	×	✓	✓	17.50	0.6859	**36.7819**	**0.9820**	**3.7951**	**0.0146**	**2.2601**	0.0087

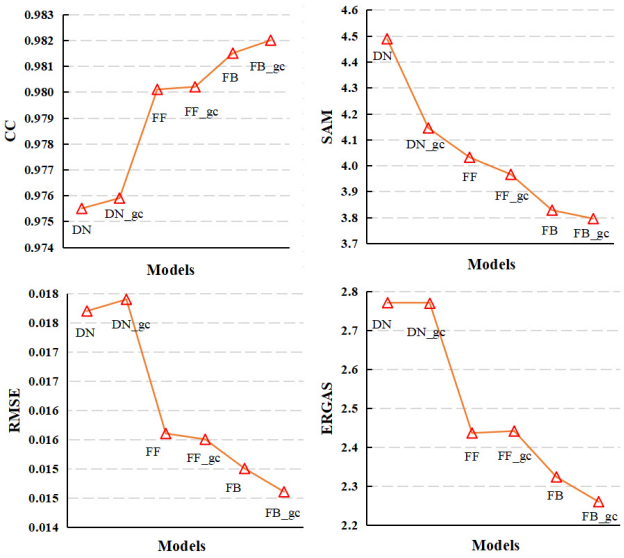


Fig. 6. Comparisons between the proposed model and its variants.

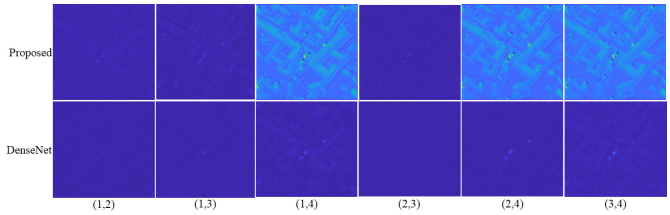


Fig. 7. Feature difference maps between the proposed model and its variant DN. (i, j) denotes the feature difference maps between the ith group and the jth group of feature maps.

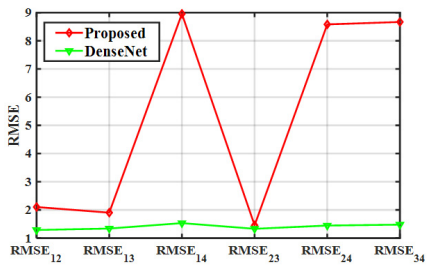


Fig. 8. RMSE comparisons between the proposed model and its variant DN. The subscript ij denotes the RMSE values between the ith group and the jth group of feature maps.

B. Competing Methods and Evaluation Methodology

To evaluate the performance of the proposed method, six widely accepted HSSR algorithms are selected, including Gram-Schmidt adaptive (GSA) [14], principal component analysis (PCA) [12], coupled nonnegative matrix factorization (CNMF) [17], HySure [18], PS-GDANet [29], UAL [27], model-guided deep convolutional network (MoG-DCN) [33], and HyperPNN [26]. These methods cover the main four categories of fusion-based HS SR methods and deep-learning-based methods. We adopt five widely used assessment metrics for evaluation, including the peak signal-to-noise ratio (PSNR), the cross correlation (CC), the spectral angle mapper (SAM), the root-mean-squared error (RMSE), and the erreur relative global adimensionnelle de synthese (ERGAS).

C. Ablation Studies

In order to verify the effectiveness of the proposed model, a set of ablation studies with different configurations is conducted to study their impact on the performance of our proposed model.

1) Model Structures: As shown in Table II, we have designed six types of model structures to study the impact of different configurations on performance improvement. We tune the parameters of all models for the best performance. We also show quantitative metrics for different models in Fig. 6 for a visual comparison. From Table II and Fig. 6, we observe that FB_gc (CFDcagaNet adopts FB mechanism and GConv) has the best performance, followed by FB (model adopts FB mechanism and conventional convolution), FF_gc (model adopts FF mechanism and GConv), FF (model adopts FF mechanism and conventional convolution), DN_gc (model adopts conventional DN and GConv), and, finally, DN (model adopts conventional DN and conventional convolution). Performance (FB > FF > DN; FB_gc > FF_gc > DN_gc) indicates that the forward guidance mechanism and FB guidance mechanism facilitate the network to transfer more efficient features compared to the original DN. Compared with the forward guidance mechanism, the FB guidance mechanism is more conducive for the network to learn discriminative characteristics from the shallow layer of the network. This is because deep features have larger receptive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

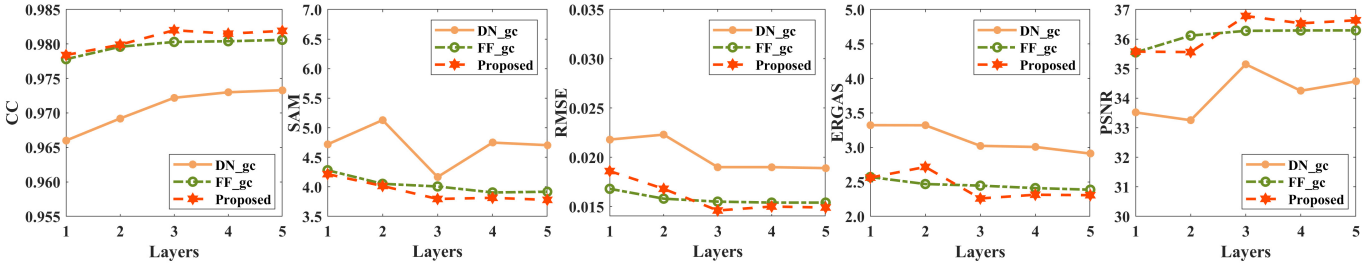


Fig. 9. Performance comparisons between different models for different layers on the Houston dataset.

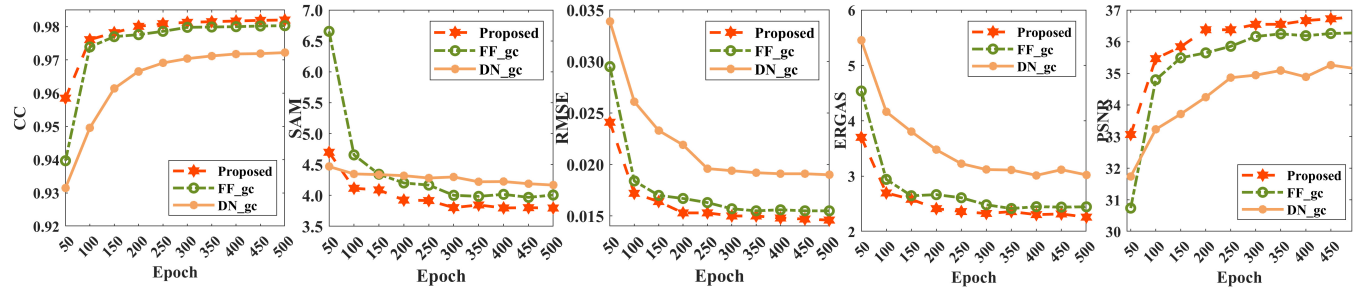


Fig. 10. Convergence with different models on the Houston dataset.

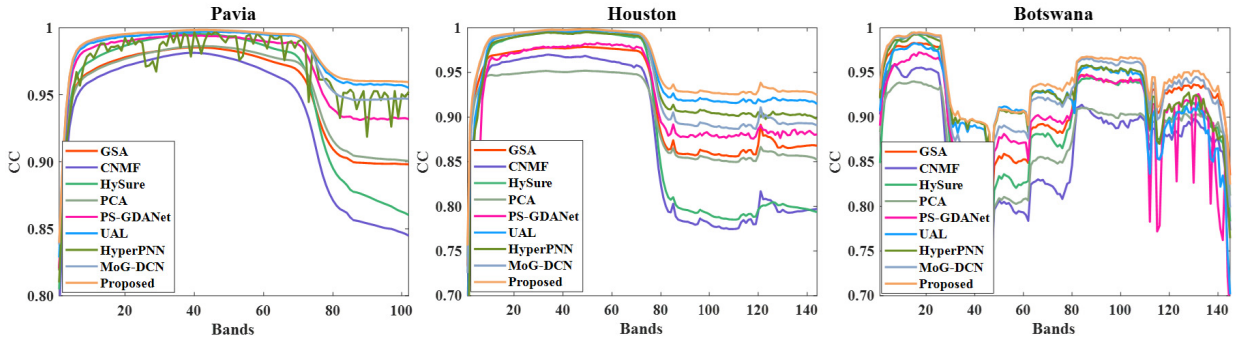


Fig. 11. CC curves as functions of the spectral bands for the competing methods on three datasets.

fields and stronger semantic information than shallow features and can provide more accurate attention guidance for shallow features.

In addition, we have also displayed the pairwise difference maps between the $i-1$ groups of reused features in the ith layer on the DN and FB models, respectively. As displayed in Fig. 7, the difference between the $i-1$ group of reused features in the ith layer in the FB model is significantly larger than that in the DN model, indicating that the proposed model can learn more discriminative features and effectively reduce redundancy.

In order to further experimentally prove that CFDcagaNet can reduce redundant features based on DN, we have compared the RMSE values between the $i-1$ group reuse features input into the ith layer and the output features of the ith layer on the DN and FB models, respectively. For the sake of fairness, we keep the other settings of the DN and FB models consistent. It can be seen from Fig. 8 that the RMSE values between the $i-1$ group reuse features input to the ith layer and the output features of the ith layer on the DN model are smaller than those on the FB model, indicating that the correlation between the input $i-1$ group reuse features and output features of the ith layer in the DN model is higher than that in the FB model. Therefore, the DN model transfers more redundant features, and the FB model yields more effective spatial–spectral transfer in the network.

2) Increasing Layers: Intuitively, the deeper the network, the better the performance. We compare the performance of the three models, DN_gc, FF_gc, and FB_gc, with different layers. The quantitative metrics CC, SAM, RMSE, ERGAS, and PSNR results are shown in Fig. 9, where we can see that better SR performance can be achieved by increasing the layers from 1 to 3. However, as we continue to increase the number of layers, the performance improvements are limited or even reduced. This is because the input image can already be finely learned by three layers of our model, and a further increase in the network depth can only increase the computational complexity.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

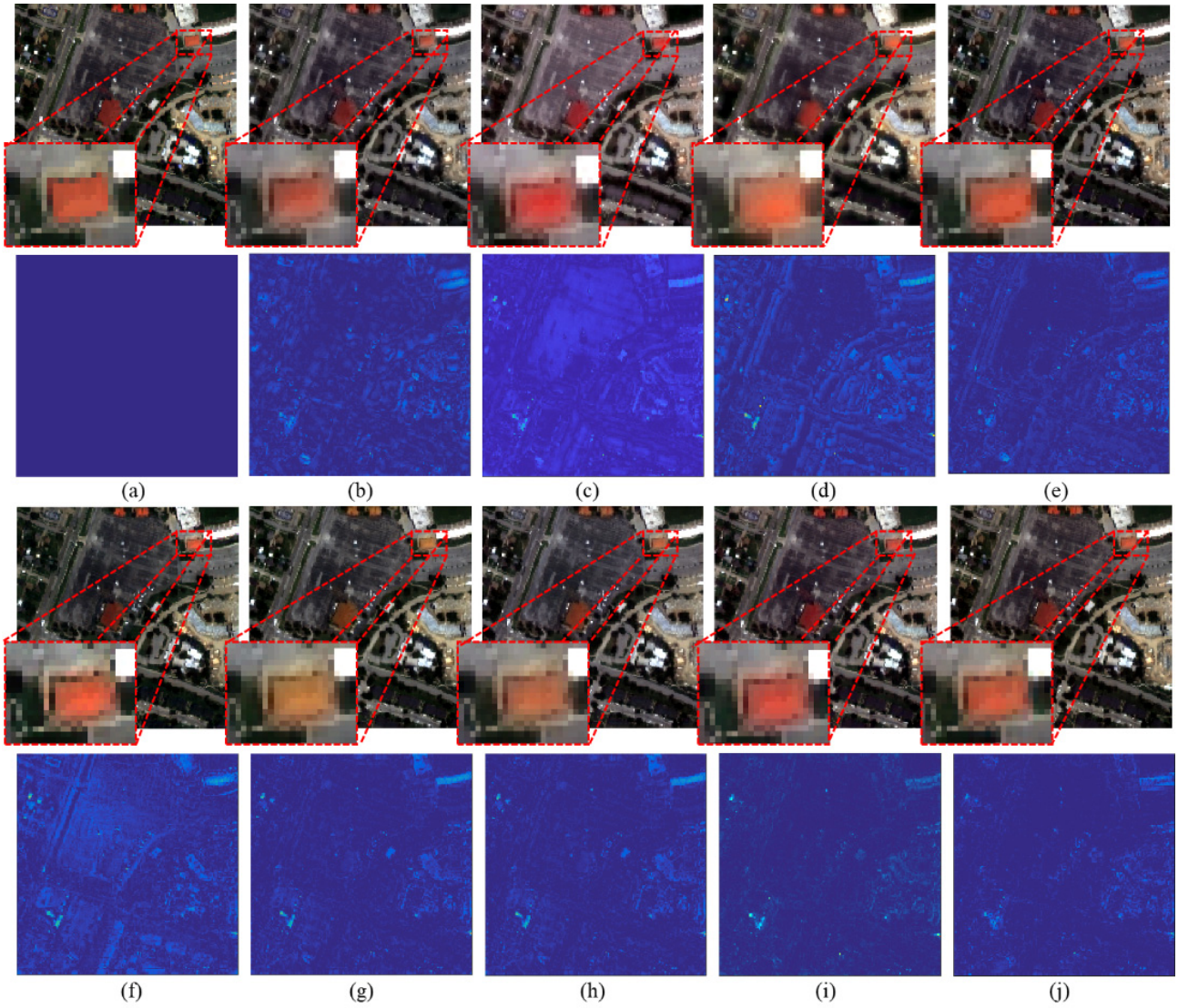


Fig. 12. Visual results and reconstruction error maps obtained by different methods on the Houston dataset. (a) Reference. (b) GSA. (c) PCA. (d) CNMF. (e) HySure. (f) PS-GDANet. (g) UAL. (h) HyperPNN. (i) MoG-GCN. (j) CFDcagaNet.

3) Convergence Rate: We also test the convergence rate of the three models, DN_gc, FF_gc, and FB_gc. As shown in Fig. 10, our model can bring a faster convergence rate. The reason is that the FB refinement mechanism adopted by our model can refine the low-level information through the high-level prior information, making the network produce a more effective transfer, thus improving the training efficiency.

4) Performance on Each Band: In order to further compare the performance of different methods, we compared the CC values of different methods in each band of three datasets, and the results are shown in Fig. 11. As can be seen, compared to the classical methods, the CNN-based methods show better performance. The proposed method shows the best performance in almost all bands of the three datasets, demonstrating the effectiveness of the proposed method in preserving spatial information.

5) Computation Complexity: Assume that the output channel number of each convolution layer is N and its kernel size is k, the number of parameters in the ith convolution layer in the standard densenet Model-DN is $(i \times N) \times N \times k \times k$. Compared with the parameters of Model-DN, the increased parameters of the proposed model mainly derive from SpaGA and SpeGA. Both SpaGA and SpeGA are composed of three layers: max-pooling, convolution, and sigmoid. The number of parameters of the convolutional layer is $1 \times 1 \times k \times k$, and the max-pooling layer and the sigmoid layer do not contain parameters. The total number of parameters of the SpaGA in

TABLE III
COMPARISON OF PARAMETERS AND FLOPS OF DIFFERENT MODELS

Method	PS-GDANet	UAL	HyperPNN	MoG-DCN	proposed
Params(M)	11.566	7.259	0.219	10.386	0.686
FLOPs(G)	49.13	337.61	5.60	584.87	17.50

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

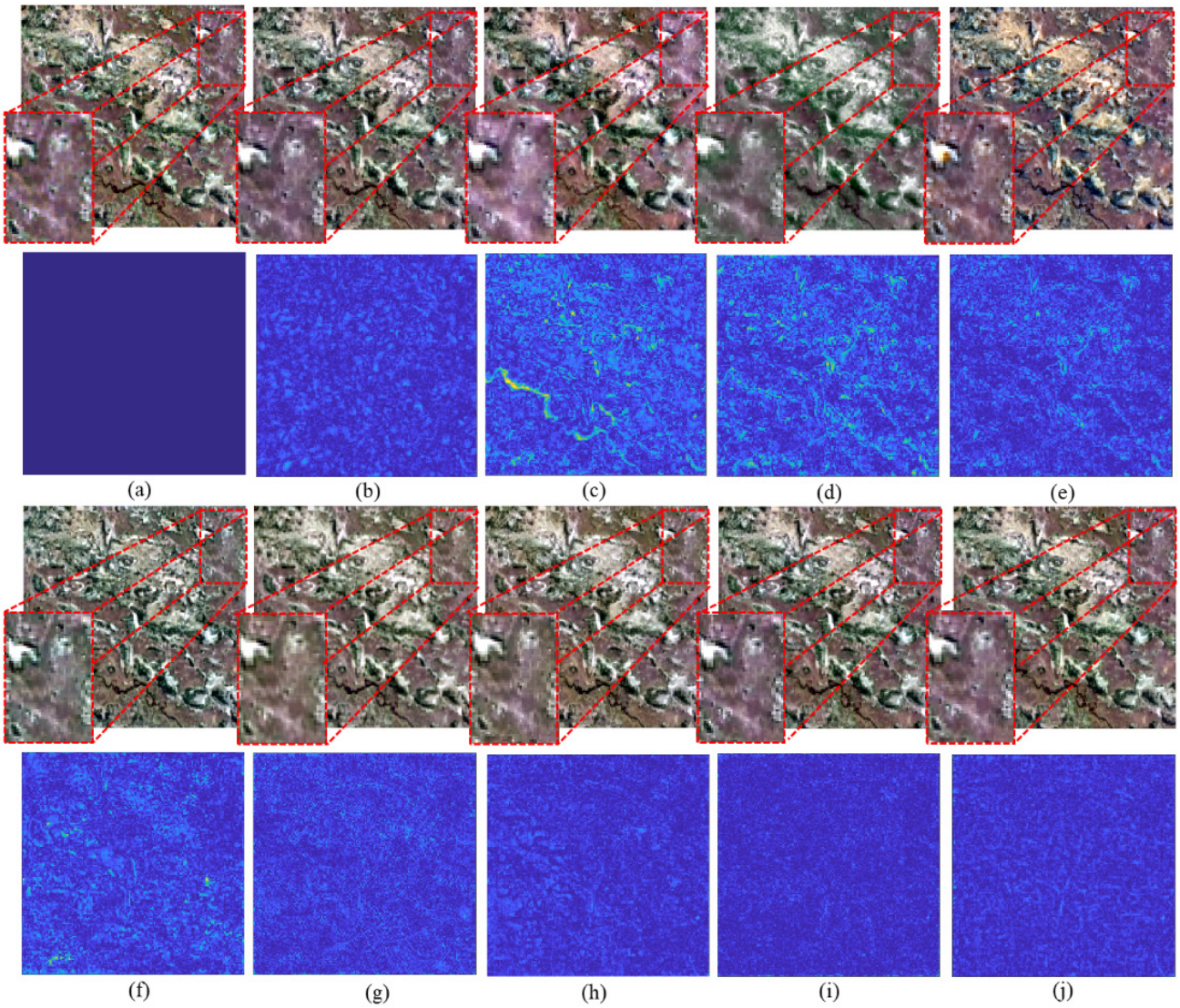


Fig. 13. Visual results and reconstruction error maps obtained by different methods on the Botswana dataset. (a) Reference. (b) GSA. (c) PCA. (d) CNMF. (e) HySure. (f) PS-GDANet. (g) UAL. (h) HyperPNN. (i) MoG-GCN. (j) CFDcagaNet.

the ith layer is $(i-1) \times 1 \times 1 \times k \times k$. Similar to SpaGA, SpeGA has $1 \times 1 \times k_c$ parameters, where k_c is the kernel size of 1-D convolution adopted in SpeGA. Thus, the total number of parameters of the SpeGA in the ith layer is $(i-1) \times 1 \times 1 \times k_c$. In the article, empirically, the output channel number of each convolution layer N and its kernel size k are set to 128 and 3, respectively. The kernel size of 1-D convolution k_c is set to seven. Specifically, the number of parameters in the ith convolutional layer of Model-DN is $i \times 147\,456$ (i.e., $(i \times N) \times N \times k \times k = (i \times 128) \times 128 \times 3 \times 3 = i \times 147\,456$), and the number of parameters in the ith layer of Model-FF and Model-FB is added $(i-1) \times 16$ (i.e., $[(i-1) \times 1 \times 1 \times k \times k] + [(i-1) \times 1 \times 1 \times k_c] = [(i-1) \times 1 \times 1 \times 3 \times 3] + [(i-1) \times 1 \times 1 \times 7] = (i-1) \times 16$) on the basis of the number of parameters in the ith layer of Model-DN. The number of added parameters $(i-1) \times 16$ is much smaller than the original parameter number $i \times 147\,456$. Table II shows that Models-FF and -FB perform better than Model-DN with only a few parameters added. As the depth of the network increases, the parameters of the convolution layer increase rapidly. In order to build an efficient network, the proposed CFDcagaNet introduces the GConv mechanism to each layer. For stability, the parameters used by the GConv are G times less than the standard convolution in DensNet. Therefore, the number of parameters can be reduced to $[(i \times N)/G] \times (N/G) \times G \times k \times k = i \times (N^2/G) \times k^2$. Comparing Model-DN and Model-FB_gc, it can be seen that the proposed CFDcagaNet (Model-FB_gc) effectively reduces the number of parameters and the running time while ensuring the network performance.

6) Comparison With State-of-the-Art Efficient CNNs: The comparison results of the proposed model and the competing deep learning-based HSSR techniques in terms of parameters and floating-point operations (FLOPs) are shown in Table III. HyperPNN is an earlier proposed network consisting of several

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
AVERAGE QUANTITATIVE RESULTS OF THE DIFFERENT METHODS ON THE PAVIA CENTER AND THE HOUSTON AND THE BOTSWANA DATASETS

Dataset	Index	GSA	PCA	CNMF	HySure	PS-GDANet	UAL	HyperPNN	MoG-DCN	CFGDNet
Pavia Center	PSNR	31.0588	29.4146	28.6376	31.2467	34.1885	36.0132	35.3829	35.5711	**36.7819**
	CC	0.9439	0.9307	0.9083	0.9504	0.9717	0.9788	0.9769	0.9786	**0.9820**
	SAM	6.4561	6.1126	6.3655	6.0739	4.6160	4.1947	4.1323	4.1243	**3.7951**
	RMSE	0.0284	0.0341	0.0375	0.0277	0.0197	0.0160	0.0173	0.0168	**0.0146**
	ERGAS	4.3052	5.1726	5.6997	3.9110	2.9457	2.5158	2.5697	2.4977	**2.2601**
Houston	PSNR	36.4650	35.1738	35.1774	36.1613	37.7148	40.4750	37.5537	39.3815	**40.6342**
	CC	0.9372	0.9305	0.9165	0.9253	0.9502	0.9676	0.9619	0.9627	**0.9725**
	SAM	4.3444	4.3798	5.2269	4.5397	4.1553	2.8794	3.3484	3.1629	**2.6835**
	RMSE	0.0150	0.0174	0.0174	0.0156	0.0130	0.0095	0.0106	0.0108	**0.0093**
	ERGAS	3.7921	4.3412	4.4009	3.9656	3.4106	2.5318	2.8294	2.7671	**2.4295**
Botswana	PSNR	39.2407	37.3970	37.3564	38.7343	39.1321	40.7817	40.8328	40.7232	**40.9147**
	CC	0.9369	0.9149	0.9075	0.9292	0.9362	0.9471	0.9520	0.9543	**0.9615**
	SAM	1.7793	2.0002	2.0702	2.0881	1.8664	1.7132	1.7029	1.6117	**1.5218**
	RMSE	0.0112	0.0139	0.0138	0.0117	0.0111	0.0092	**0.0091**	0.0093	**0.0091**
	ERGAS	1.6347	1.9591	2.0752	1.7470	1.7687	1.6371	2.2075	1.4054	**1.3710**

simple convolutional layers, so it requires fewer parameters and FLOPs. UAL contains multiple convolutional layers, and most of the convolutional layers have been calculated between two stages or three branches many times, so it requires a large number of FLOPs. However, due to the shared parameters in the network, the number of parameters in the network does not increase rapidly with FLOPs. Similarly, MoG-DCN contains multistage iterative operations, so it requires a high number of FLOPs, and its parameter amount is acceptable due to parameter sharing. From Table III, we observe that the proposed model achieves better performance than other state-of-the-art CNN architectures with acceptable parameters and FLOPs.

D. Comparative Experiments

In this section, we provide some subjective and objective results of the proposed method and other competing methods on the Houston dataset, the Pavia Center dataset, and the Botswana dataset to verify the performance of the proposed CFDcagaNet.

The pseudocolor and the reconstruction error maps on the Houston dataset obtained by the competing methods are shown in Fig. 12. Since human is not sensitive to small differences, we also calculate the reconstruction error between the pansharpened results and ground-truth image for better subjective evaluation. For a better comparison, a rectangular area is shown in the lower right of each fused image, which is magnified three times for small objects to evaluate the effectiveness of the proposed method. We can observe that the GSA and PCA methods introduce serious spectral distortion in most urban areas. This is because the PAN image is simply modeled as the weighted average of the HS band in GSA and PCA methods, which leads to spectral distortion. Fig. 12(d) and (e) shows that the CNMF and HySure methods also produce blurring artifacts on the Houston dataset. Compared to the classical methods, the deep-learning-based methods cause smaller pixel differences for most areas. This is because CNNs have strong feature learning ability and can fully learn the details from LR-HS images. In contrast, the product of CFDcagaNet maintains the highest fidelity to the ground-truth image, which indicates the superiority in learning the spatial and spectral characteristics of the CFDcagaNet.

We present the fused HS images of the competing methods on the Botswana dataset and its corresponding reconstruction error maps in Fig. 13. As can be seen, the GSA, PCA, CNMF, and HySure all suffer from severe spectral and spatial distortions [see the closeup views of Fig. 13(b)–(e)], as reconstruction errors of their SR results are obvious. By contrast, the CNN-based PS-GDANet, HyperPNN, MoG-GCN, and CFNcagaNet can create satisfactory results; especially, the fusion result of our CFNcagaNet method is the closest to the reference result. This is because the learning process in CFNcagaNet is simplified by introducing the cross-feedback refinement mechanism. The network can well learn effective features from the low-level layers.

Fig. 14 exhibits the results in pseudocolor of different methods on the Pavia Center dataset. From Fig. 14, we can see that GSA and PCA yield poor results with severe spectral distortion [see the closeup views of Fig. 14(b) and (c)]. CNMF and HySure show obvious differences at the boundary of the ground feature [see the closeup views of Fig. 14(d) and (e)], indicating that these methods do not work well in enhancing spatial details. In contrast, deep learning-based PS-GDANet, UAL, HyperPNN, MoG-GCN, and CFDcagaNet can achieve better performance in spatial detail injection and spectral fidelity than classical methods. The proposed CFDcagaNet

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

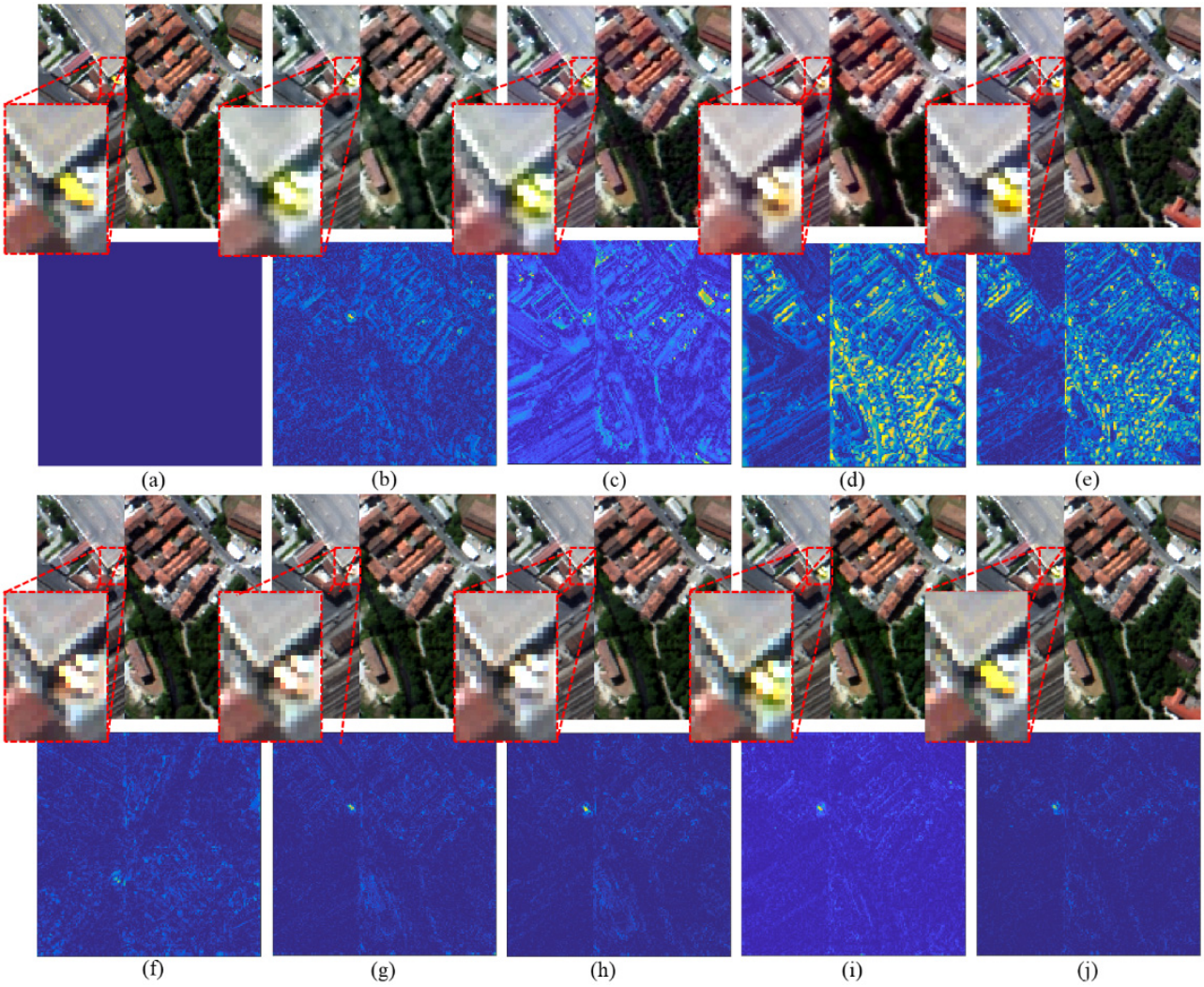


Fig. 14. Visual results and reconstruction error maps obtained by different methods on the Pavia Center dataset. (a) Reference. (b) GSA. (c) PCA. (d) CNMF. (e) HySure. (f) PS-GDANet. (g) UAL. (h) HyperPNN. (i) MoG-GCN. (j) CFDcagaNet.

exhibits more dark blue areas, indicating that our CFDcagaNet causes the smallest spatial–spectral differences for most areas.

Table IV reports the quantitative evaluation results of different methods on the three datasets. The limited performance of classical methods indicates they may lack robustness to different datasets. It can be seen that the deep learning-based methods i.e., PS-GDANet, UAL, HyperPNN, MoG-GCN, and CFDcagaNet, provide better reconstruction results than other classical approaches, mainly by virtue of their strong feature learning ability and highly nonlinear mapping. Particularly, the proposed CFDcagaNet achieves the most outstanding performance in terms of four major metrics, i.e., PSNR, CC, SAM, RMSE, and ERGAS. The proposed method can achieve good results mainly due to the following reasons.

1) The proposed model has the ability to make full use of all the hierarchical features within and in between two branches.
2) An effective cross-feedback refinement mechanism is developed to fully exploit the high-level posterior information to refine the low-level features in a FB manner across two branches, yielding more effective spatial–spectral transfer in the network.

IV. Conclusion

This study investigated a cross-feedback dense network, CFDcagaNet, with applications to the fusion-based HRSR. Most existing dense network based SR models fully connect adjacent layers in a FF manner, inevitably introducing redundancy. Instead, our proposed CFDcagaNet infers attention maps from the high-layer features of two branches, and then, the attention maps are multiplied to the low-layer features in a FB manner across two branches. As a result, CFDcagaNet is capable of learning discriminative characteristics from a low layer of the network, and all the high-to-low layer representations are spatially more accurate and semantically richer. Extensive experiments conducted on widely used datasets demonstrate the superiority of the CFDcagaNet in improving spatial resolution and spectral fidelity of HS images.

References

[1] H. Su, W. Yao, Z. Wu, P. Zheng, and Q. Du, "Kernel low-rank representation with elastic net for China coastal wetland land cover classification using GF-5 hyperspectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 171, pp. 238–252, Jan. 2021.
[2] Y. Dong, Y. Jin, and S. Cheng, "Clustered multiple manifold metric learning for hyperspectral image dimensionality reduction and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.

[3] S. Mei, R. Jiang, X. Li, and Q. Du, "Spatial and spectral joint super-resolution using convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4590–4603, Jul. 2020.

[4] J. Hu, X. Jia, Y. Li, G. He, and M. Zhao, "Hyperspectral image super-resolution via intrafusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7459–7471, Oct. 2020.

[5] J. T. Peng, W. Sun, L. Ma, and Q. Du, "Discriminative transfer joint matching for domain adaptation in hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 972–976, Jun. 2019.

[6] W. Sun, W. Shao, J. Peng, G. Yang, X. Meng, and Q. Du, "Multiscale low-rank spatial features for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2020.

[7] Y. Dong, Y. Zhang, and B. Du, "Symmetric information–theoretic metric learning for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1470–1480, 2022.

[8] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[9] J. Qu, Q. Du, Y. Li, L. Tian, and H. Xia, "Anomaly detection in hyperspectral imagery based on Gaussian mixture model," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9504–9517, Nov. 2021.

[10] Y. Qu, H. Qi, B. Ayhan, C. Kwan, and R. Kidd, "DOES multispectral/hyperspectral pansharpening improve the performance of anomaly detection?" in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 6130–6133.

[11] L. Loncan *et al.*, "Hyperspectral pansharpening: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[12] P. S. Chavez and A. Y. Kwarteng, "Extracting spectral contrast in landsat thematic mapper image data using selective principal component analysis," *Photogramm. Eng. Remote Sens.*, vol. 55, no. 3, pp. 339–348, 1989.

[13] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[14] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS+Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[15] J. G. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, Nov. 2000.

[16] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and PAN imagery," *Photogramm. Eng. Remote Sens.*, vol. 72, no. 5, pp. 591–596, May 2006.

[17] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[18] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[19] Q. Wei, N. Dobigeon, and J. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[20] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J. Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[21] W. Sun, K. Ren, X. Meng, C. Xiao, G. Yang, and J. Peng, "A band divide-and-conquer multispectral and hyperspectral image fusion method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[22] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, early access, Apr. 13, 2022, doi: 10.1109/MGRS.2022.3145854.

[23] Y. Wei, Q. Yuan, H. Shen, and L. Zhang, "Boosting the accuracy of multispectral image pansharpening by learning a deep residual network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1795–1799, Oct. 2017.

[24] M. Jiang, H. Shen, J. Li, Q. Yuan, and L. Zhang, "A differential information residual convolutional neural network for pansharpening," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 257–271, Jun. 2020.

[25] Q. Liu *et al.*, "Hybrid attention based residual network for pansharpening," *Remote Sens.*, vol. 13, no. 10, p. 1962, May 2021.

[26] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[27] L. Zhang, J. Nie, W. Wei, Y. Zhang, S. Liao, and L. Shao, "Unsupervised adaptation learning for hyperspectral imagery super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3070–3079.

[28] W. Dong, T. Zhang, J. Qu, S. Xiao, J. Liang, and Y. Li, "Laplacian pyramid dense network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[29] W. Dong, S. Hou, S. Xiao, J. Qu, Q. Du, and Y. Li, "Generative dual-adversarial network with spectral fidelity and spatial enhancement for hyperspectral pansharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 10, 2021, doi: 10.1109/TNNLS.2021.3084745.

[30] W. Wang, E. Xie, P. Sun, W. Wang, and C. Shen, "TextSR: Content-aware text super-resolution guided by recognition," 2019, *arXiv:1909.07113*. [Online]. Available: https://arxiv.org/abs/1909.07113

[31] J. He, J. Li, Q. Yuan, H. Shen, and L. Zhang, "Spectral response function-guided deep optimization-driven network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 18, 2021, doi: 10.1109/TNNLS.2021.3056181.

[32] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C.-W. Chan, and W. Philips, "Spatial-spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[33] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[34] J. Ma, W. Yu, C. Chen, P. Liang, X. Guo, and J. Jiang, "Pan-GAN: An unsupervised pan-sharpening method for remote sensing image fusion," *Inf. Fusion*, vol. 62, pp. 110–120, Oct. 2020.

[35] Y. Qu, R. K. Baghbaderani, H. Qi, and C. Kwan, "Unsupervised pansharpening based on self-attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3192–3208, Apr. 2021.

[36] Y. Qu, H. Qi, C. Kwan, N. Yokoya, and J. Chanussot, "Unsupervised and unregistered hyperspectral image super-resolution with mutual Dirichlet-Net," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[37] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.

[38] K. Jiang, Z. Wang, P. Yi, T. Lu, J. Jiang, and Z. Xiong, "Dual-path deep fusion network for face image hallucination," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 1, pp. 378–391, Jan. 2022.

[39] Y. Shi, H. Zhong, Z. Yang, X. Yang, and L. Lin, "DDet: Dual-path dynamic enhancement network for real-world image super-resolution," *IEEE Signal Process. Lett.*, vol. 27, pp. 481–485, 2020.

[40] G. Huang, S. Liu, L. V. D. Maaten, and K. Q. Weinberger, "CondenseNet: An efficient DenseNet using learned group convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 2752–2761.

[41] H. Zhao, O. Gallo, I. Frosio, and J. Kautz, "Loss functions for image restoration with neural networks," *IEEE Trans. Comput. Imag.*, vol. 3, no. 1, pp. 47–57, Mar. 2017.

[42] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.



Wenqian Dong (Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in June 2020.

She has been a Visiting Scholar with Simon Fraser University, Burnaby, BC, Canada, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published more than 20 papers in refereed journals and conferences. Her research interests include compressed sensing, machine learning, and hyperspectral remote sensing image processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.



Jiahui Qu (Member, IEEE) received the B.S. degree in communication engineering from Yantai University, Yantai, China, in 2014, and the Ph.D. degree in communication and information systems, Xidian University, Xi'an, China, in June 2020.

She was an Exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA, from 2018 to 2019. She is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, Xidian University. She has published over 20 papers in academic journals and conferences, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS. Her research interests include remote sensing image detection, image fusion, image super-resolution, and deep learning.



Tongzhen Zhang received the B.E. degree in intelligence science and technology from Xidian University, Xi'an, China, in 2019, where he is currently pursuing the M.S. degree with the Image Coding and Processing Center, State Key Laboratory of Integrated Service Network.

His research interests include hyperspectral image fusion and deep learning.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

He joined the School of Telecommunications Engineering, Xidian University, in 1999, where he is currently a Professor. He is also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks, Xidian University. His research interests include image and video processing, hyperspectral image processing, and high-performance computing.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland Baltimore County, Baltimore, MD, USA, in 2000.

She is currently a Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a Fellow of the International Society for Optics and Photonics (SPIE). She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS). She served as the Co-Chair of the Data Fusion Technical Committee of the IEEE GRSS from 2009 to 2013 and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She is the General Chair for the 4th IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing in Shanghai, China, in 2012. She served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:51:15 UTC from IEEE Xplore. Restrictions apply.